

Discovering Structure

Semantic Spaces and Unsupervised Modeling

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Introduction to NLP — MSc. DH EdC-PSL

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1 Recap

1.1 Language Model

1.1.1 The Language Modeling Problem



Language Model

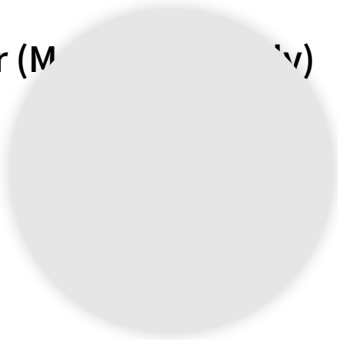
A Language Model (LM) estimates the probability of pieces of text. Given a sequence of text $(w_1, w_2, \dots, w_S)$, it answers the question:

What is $P(w_1, w_2, \dots, w_S)$?

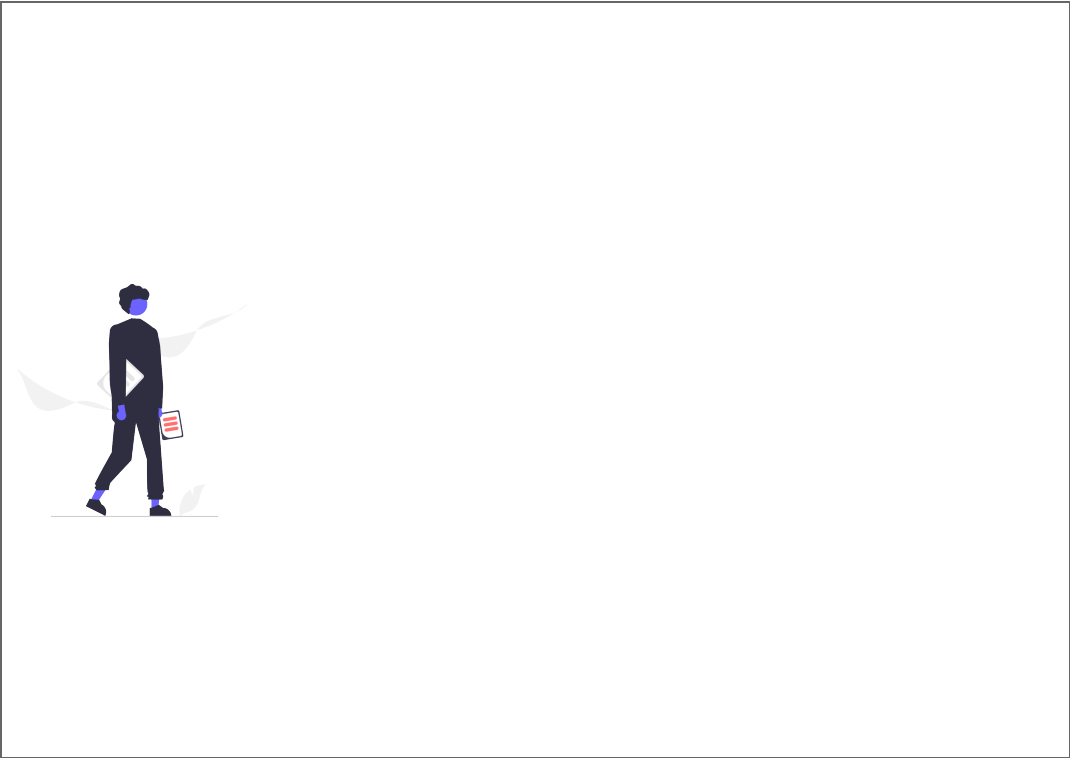
How to compute P ?

- Count-based (count): n -gram
 - Neural method (learn): Transformer
 - Masked Language Modeling (encoder-only, BERT)
 - Causal Language Modeling (decoder-only, GPT)
- $\rightarrow$ encode:
- grammaticality
 - semantic plausability
 - stylistic consistency
 - knowledge (?)

1.1.2 Transformer (MVA)



Transformer = Support Groups



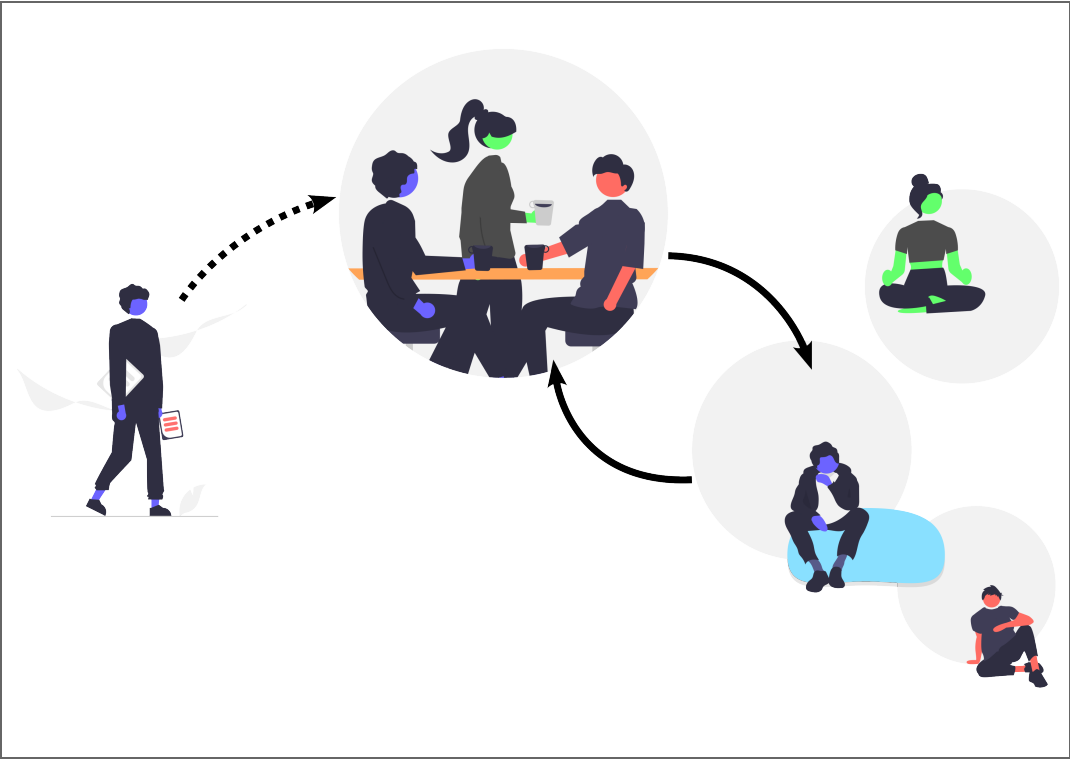
Transformer = Support Groups



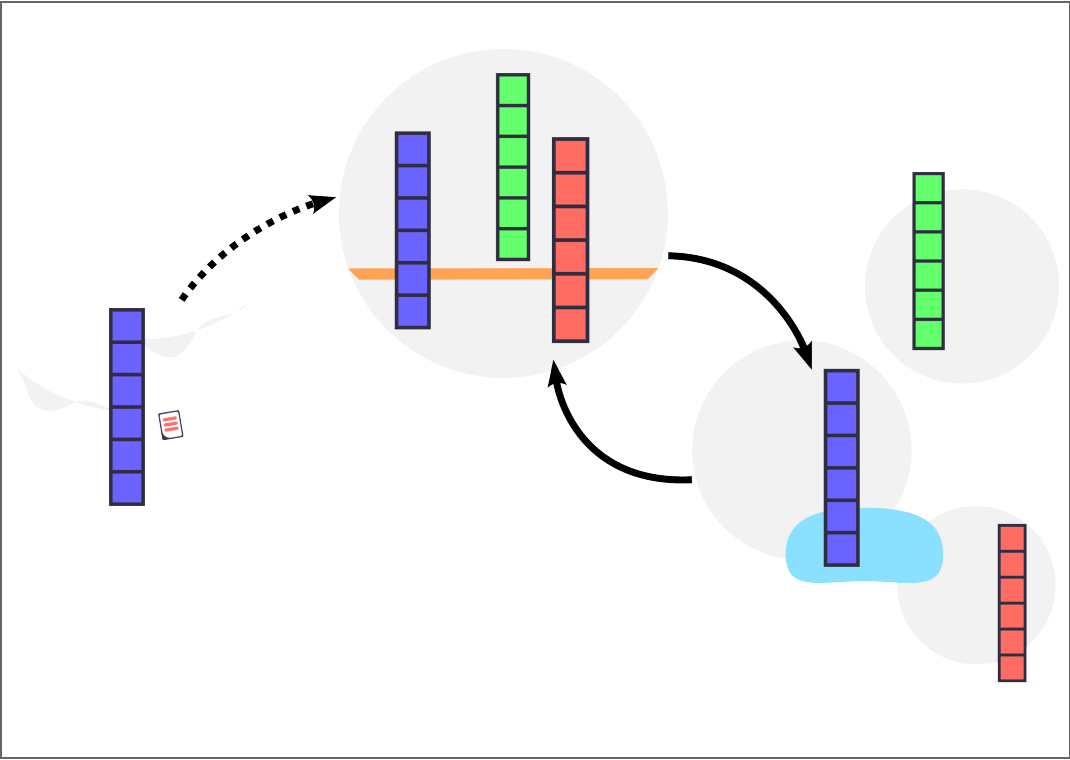
Transformer = Support Groups



Transformer = Support Groups



Transformer = Support Groups



1.2 Pre-trained LM

1.2.1 Word Embeddings — Feature Extraction

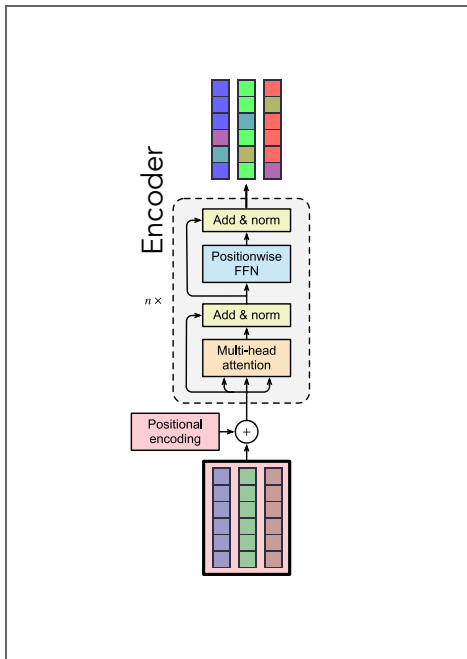


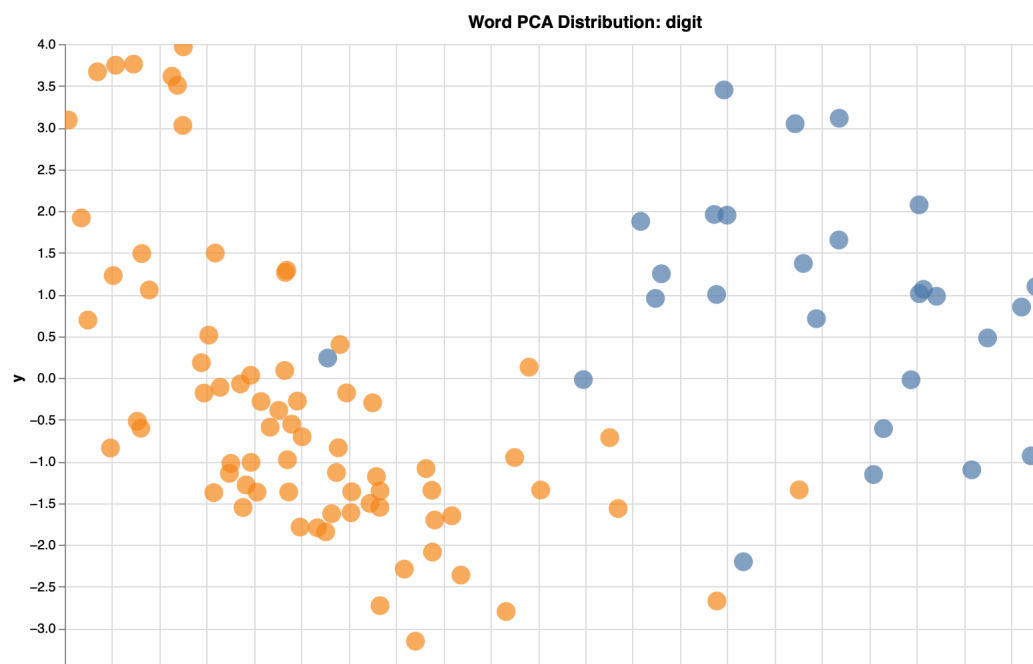
Diagram encoder-only model (e.g.; BERT).

- Tokenize text sequence
 - BERT means Bidirectional Encoder Representations from Transformers.
 - [CLS] bert means bid ##ire ##ction ##al
en##code ##r representations from transformers
.[SEP]
- Run through the pre-trained LM
 - Output embeddings: $(N_{\text{batch}} \times N_{\text{tokens}} \times d_{\text{model}})$
- Extract contextual embedding at the position of choice

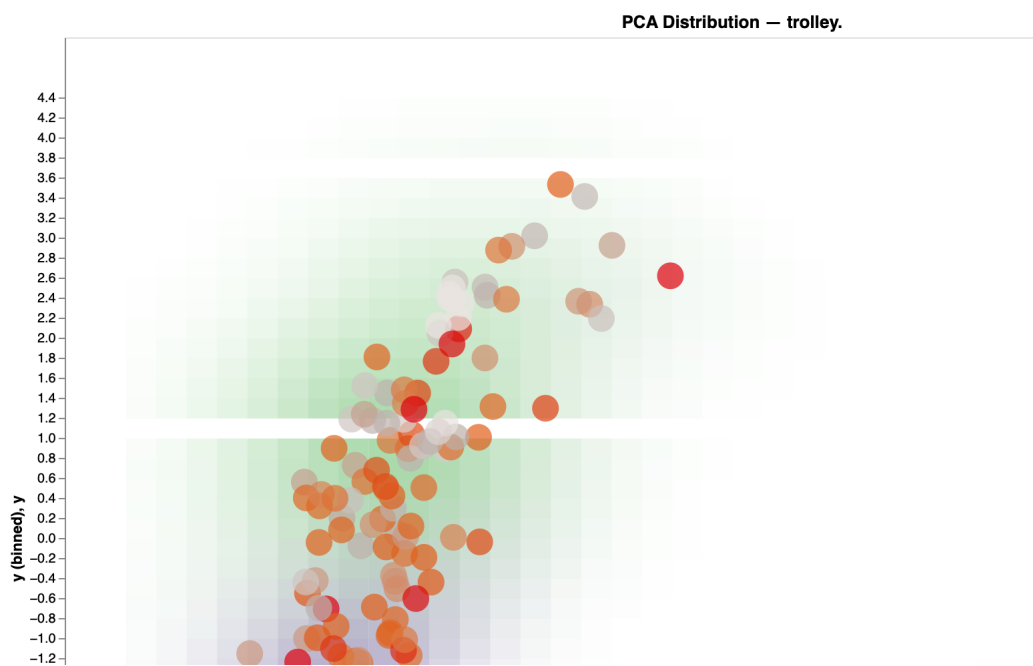
1.2.2 In practice

```
1 from transformers import XXModelForTask, XXTokenizer
2
3 # load the LM
4 model_name = "<name_of_the_model>"
5 model = XXModelForTask.from_pretrained(model_name)
6 tokenizer = XXTokenizer.from_pretrained(model_name)
7
8 # load your data
9 text = "Bla bla bla" # some text you are interested in
10
11 # tokenize the text(s)
12 tokenized_text = tokenizer(text, return_tensors="pt")
13
14 # process it with the model (+ do downstream analysis)
15 output = model(**tokenized_text)
```

1.2.2.1 WORD SENSE DISAMBIGUATION



1.2.2.2 SEMANTIC SHIFT EXPLORATION



2 From Words to Document Representations

2.1 Introduction

2.1.1 Definitions



Representation / Embedding

Numeric vector representation that a machine can understand.

Symbolic Language $\rightarrow$ LM $\rightarrow$ Representation

Document = sequence of words ($\rightarrow$ tokens)

– depending on context and applications (sentence, whole document (short/long), chunks, ...)



2.2 Count-based Approach

2.2.1 Bag-of-Words (BoW)

Document = **multiset** of the **words** it contains.

Let V be the vocabulary (based on corpus), BoW represents documents as vectors of dimension $|V|$, where each component is associated with a token.

- Main variants: binary, **raw count**, (frequency)



BoW Illustration of the first chapter of L'Etranger (Camus, 1942); top 150 words.

2.2.2 TF-IDF



Highlight Relative Importance

Weight the terms according to their specificity across documents in the corpus, based on *term specificity* (Sparck Jones 1972).

$$\mathrm{TFIDF}(w_i, t) = f_{t,d} \cdot \log\left(\frac{N}{n_t}\right)$$

- Term Frequency: $f_{t,d}$
- Inverse Document Frequency: $\log\left(\frac{N}{n_t}\right)$



IDF Variants

There exists many variants of the TFIDF formula. For instance, by default, while gensim applies the formula above (with $\log_2$), sklearn applies: $\mathrm{IDF} = \log\left(\frac{1+N}{1+n_t}\right) + 1$. (And both perform L2-normalization afterwards).

These variants can be configured with parameters of the functions.

Word	df	idf
Romeo	1	1.57
salad	2	1.27
Falstaff	4	0.966
forest	12	0.489
battle	21	0.246
wit	34	0.037
fool	36	0.012
good	37	0
sweet	37	0

Document Frequency (df) and Inverse Document Frequency (idf) for some words in Shakespeare's 37 plays.

2.2.3 Limitations

- Context-insensitive
 - No word order (“*The pig and I ate a salad*” = “*I ate the pig and a salad*”)
 - Cannot handle polysemy (“*The match was fire, it was lit!*” ~ “*The fire was lit by a match*”)
- High-dimensional sparse representation
 - Vocabulary can rapidly grow (easily $(|V| > 10^5)$) (cf. *curse of dimensionality*)
 - Orthogonality — no concept of semantic similarity at the word-level (e.g.; “*The cat sat on the mat*” ≠ “*A feline rested on the carpet*”)

2.2.4 Quick Example

Find the binary and absolute frequencies BoW and TF-IDF representations for:

- $\backslash(d_1\backslash)$: The final match was fire.
- $\backslash(d_2\backslash)$: The fire was started by a match and the fire spread quickly.

<u>Voc.</u>	<u>Binary BoW</u>	<u>Raw Count BoW</u>
<u>TF-IDF</u>		

		d_1	d_2
a	→	☐	☐
and	→	☐	☐
by	→	☐	☐
final	→	☐	☐
fire	→	☐	☐
match	→	☐	☐
quickly	→	☐	☐
spread	→	☐	☐
started	→	☐	☐
the	→	☐	☐
was	→	☐	☐

		d_1	d_2
a	→	0	1
and	→	0	1
by	→	0	1
final	→	1	0
fire	→	1	1
match	→	1	1
quickly	→	0	1
spread	→	0	1
started	→	0	1
the	→	1	1
was	→	1	1

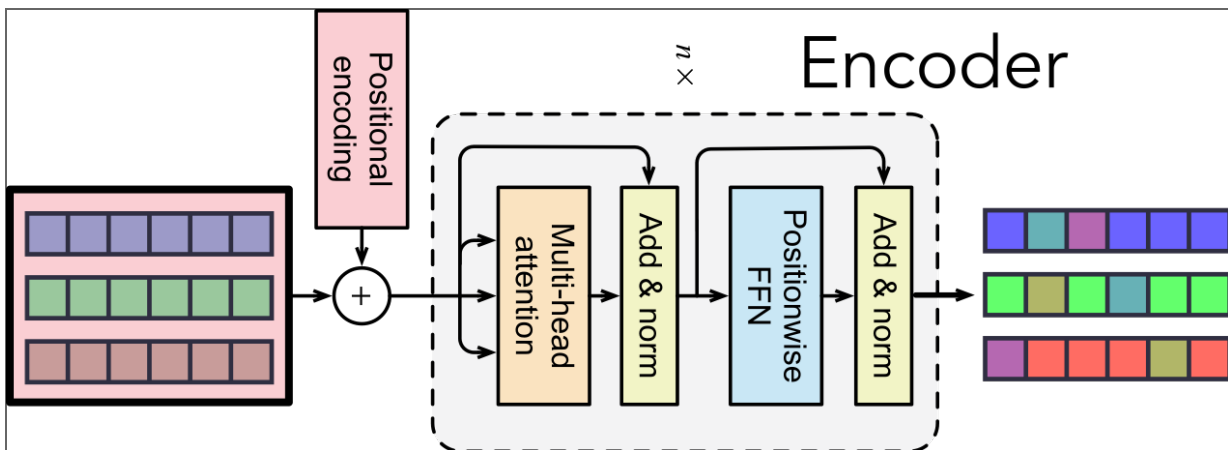
		d_1	d_2
a	→	0	1
and	→	0	1
by	→	0	1
final	→	1	0
fire	→	1	2
match	→	1	1
quickly	→	0	1
spread	→	0	1
started	→	0	1
the	→	1	2
was	→	1	1

TF $\times$ IDF, with $IDF = \log(N/n_t)$

- for words with $(n_t=1) \rightarrow \log(2) \approx 0.693$
- for words with $(n_t=2=N) \rightarrow \log(1)=0$

2.3 Transformer-based

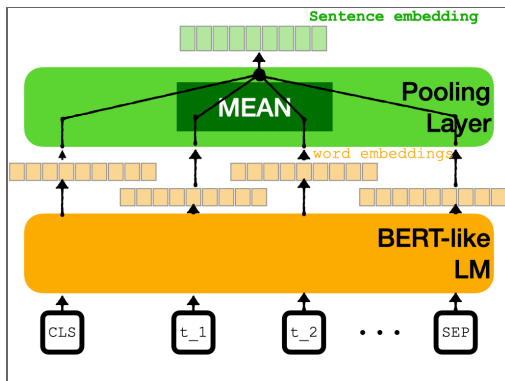
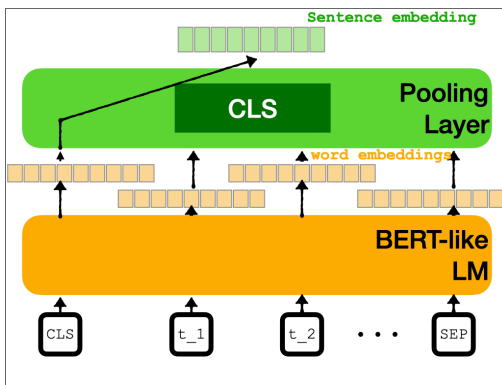
2.3.1 Reminder: Contextual Word Embedding



Encoder-based feature extractor diagram (token-level).

How to have representations for the whole document?

2.3.2 Naive Pooling Strategies



works to some extent but results are not that good, & unfit for large-scale semantic comparison...

BERT was not trained for sentence-level representation!

2.3.3 SentenceBERT

Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (Reimers and Gurevych 2019)

Nils Reimers and Iryna Gurevych

Ubiquitous Knowledge Processing Lab (UKP-TUDA)

Department of Computer Science, Technische Universität Darmstadt

www.ukp.tu-darmstadt.de

Abstract

BERT (Devlin et al., 2018) and RoBERTa (Liu et al., 2019) has set a new state-of-the-art performance on sentence-pair regression tasks like semantic textual similarity (STS). However, it requires that both sentences are fed into the network, which causes a massive computational overhead: Finding the most similar pair in a collection of 10,000 sentences requires about 50 million inference computations (~65 hours) with BERT. The construction of BERT makes it unsuitable for semantic similarity search as well as for unsupervised tasks like clustering.

In this publication, we present Sentence-BERT (SBERT), a modification of the pretrained

tic similarity comparison, clustering, and information retrieval via semantic search.

BERT set new state-of-the-art performance on various sentence classification and sentence-pair regression tasks. BERT uses a cross-encoder: Two sentences are passed to the transformer network and the target value is predicted. However, this setup is unsuitable for various pair regression tasks due to too many possible combinations. Finding in a collection of $n = 10\,000$ sentences the pair with the highest similarity requires with BERT $n \cdot (n-1)/2 = 49\,995\,000$ inference computations. On a modern V100 GPU, this requires about 65 hours. Similar, finding which of the over 40 million existent questions of Quora is the most similar

2.3.4 SBERT="BERT+pooling"

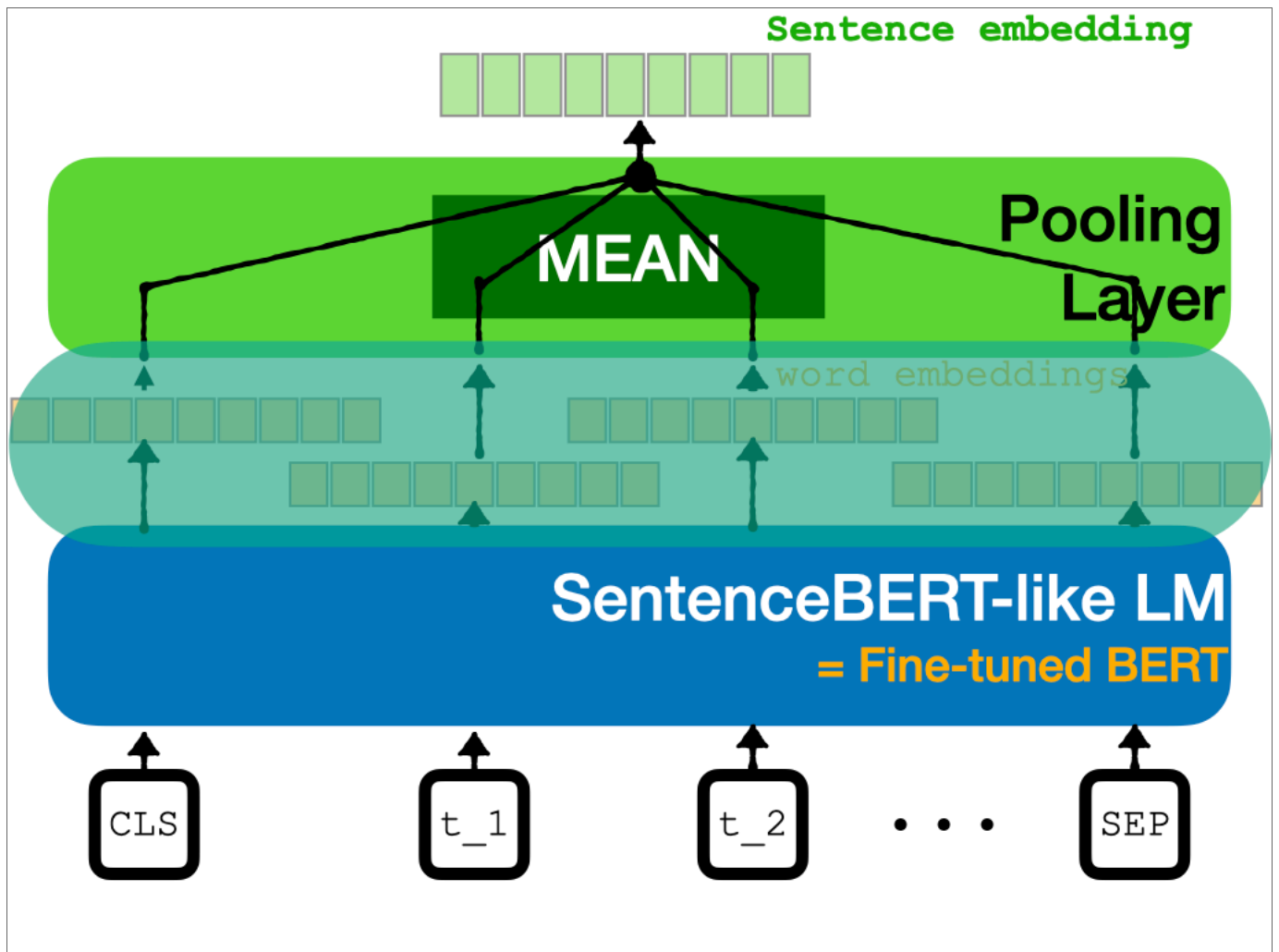
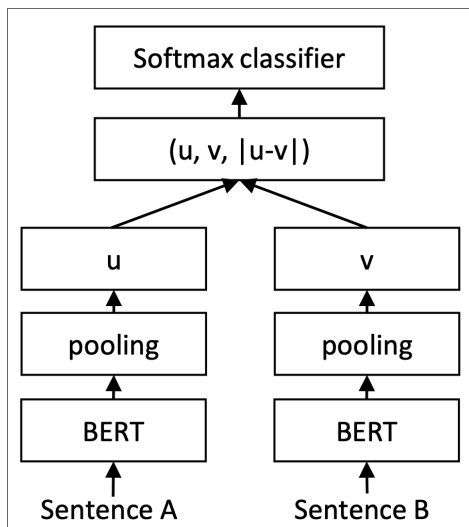


Illustration of SentenceBERT-like models architecture at inference time.

2.3.5 Training Principle



SBERT architecture with classification objective function. Fig. 1 extracted from (Reimers and Gurevych 2019).

- Siamese network
 - Shared weights
- Aggregation of pooled embeddings
 - CLS, MEAN, MAX
- Three objective functions (sequence-level)
 - Classification (**NLI** used to **train** the model)
 - Regression
 - Triplet

2.4 Summary

2.4.1 Key differences

	TF-IDF	SentenceBERT
Approach	Count-based	Neural
Vector	Sparse ($d \sim 10^{5-6}$)	Dense ($d \sim 10^{2-3}$)
Interpretability	✓	✗
Corpus-dependent	Yes (relative)	No (absolute)
Context	✗	✓
Semantic	✗	✓

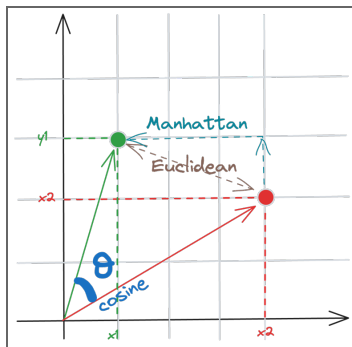
3 Geometrical Spaces

3.1 Notions of distances

3.1.1 Geometrical Distances in 2 dimensions

Different notions of distance, e.g.:

- **Euclidean:** $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
- **Manhattan (L1):** $|x_1 - x_2| + |y_1 - y_2|$
- **Cosine:** $1 - \frac{x_1 y_1 + x_2 y_2}{\sqrt{x_1^2 + x_2^2} \sqrt{y_1^2 + y_2^2}}$
 - ⚠ formally cosine distance is not a metric, but a dissimilarity measure



2D diagram.

3.1.2 Geometrical Distances in N dimensions

Different notions of distance, e.g.:

- **Euclidean:** $d(x,y) = \sqrt{\sum_{i=1}^N (x_i - y_i)^2}$
- **Manhattan (L1):** $d(x,y) = \sum_{i=1}^N |x_i - y_i|$
- **Cosine:** $1 - \cos\theta = 1 - \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$
 - ⚠️ formally not a metric, but a dissimilarity measure



Mathematical Formalization

1. The distance between an object and itself is always zero.
2. The distance between distinct objects is always positive.
3. Distance is symmetric.
4. Distance satisfies the triangle inequality.



3.1.3 Beware of the curse!



Curse of dimensionality (*coined by (Bellman 1966)*) (see e.g.: ([Domingos 2012](#)))

- **Sparsity:** data points occupy a tiny fraction of the space.
- **Distance collapse:** distances between points converge → “everything is equally far.”
- **Implication:** similarity measures lose discriminative power.

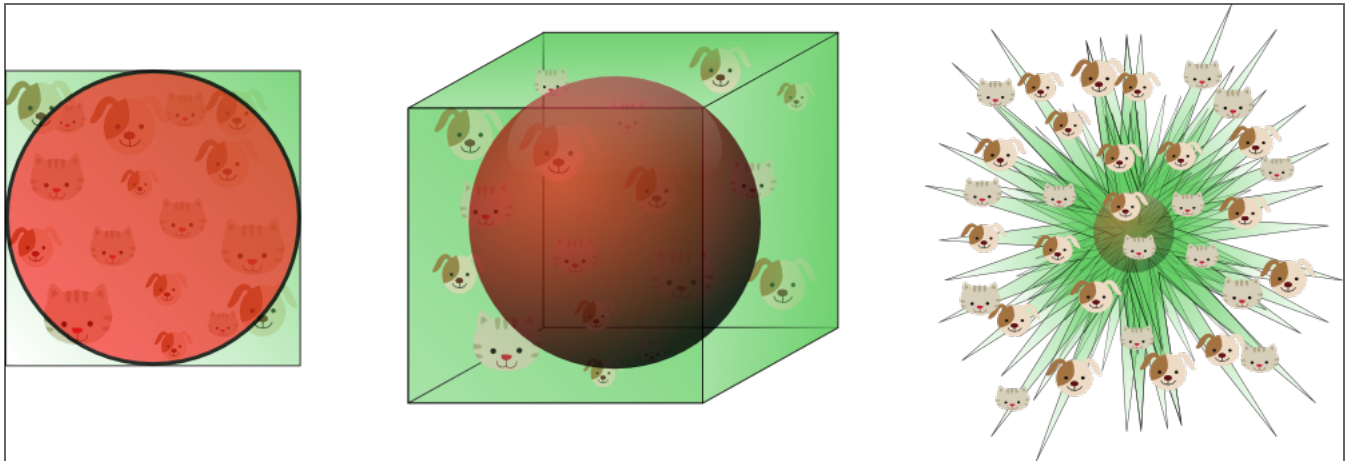


Illustration of the curse of dimensionality: As dimensionality increases, a larger percentage of the data resides at the border of the feature space. Extracted from ([Spruyt 2015](#)).

3.1.4 In practice: (most often) cosine similarity

Dot Product: $\mathbf{A} \cdot \mathbf{B} =$

$\|\mathbf{A}\| \|\mathbf{B}\| \cos\{\theta\}$

Cosine similarity: $\cos\{\theta\} =$

$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$

$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$



Origins

Use of cosine similarity for related tasks is inherited from Information Retrieval and Vector Space Models (Salton, Wong, and Yang 1975).



No Free Lunch

Mathematically also suffers from the curse of dimensionality, but **empirically** works better in practice (most often)...

... still has known limitations, e.g.:

- can sometimes yield arbitrary (and therefore meaningless) 'similarities' (Steck, Ekanadham, and Kallus 2024);
- relative to human judgements, cosine underestimates similarity of frequent words (Zhou et al. 2022).

3.2 Dimensionality Reduction

3.2.1 From high- to lower-dimensional spaces

Can serve different purposes:

- Mitigate curse of dimensionality
- Visualisation & Exploration
- Analytical

Reminder (?): various methods exist (MDS, PCA, tSNE, UMAP, ...)

3.2.2 Quick Cheatsheet

MDS (Multidimensional Scaling)

- Preserves pairwise **distances**.
- Good for small datasets, metric interpretability.
- Computationally expensive (to untractable) for large datasets.

Key params: metric, n_init, max_iter

t-SNE (t-Distributed Stochastic Neighbor Embedding)

- Preserves **local neighborhoods**.
- Excellent for cluster visualization.
- Nonlinear, stochastic, slower.

Key params: perplexity, learning_rate, n_iter

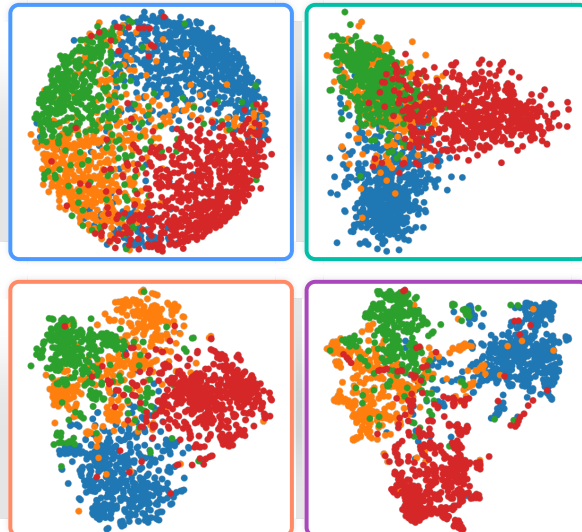
PCA (Principal Component Analysis)

- Linear projection maximizing **variance**.
- Fast, deterministic, interpretable axes.
- Captures global but not nonlinear structure.

UMAP (Uniform Manifold Approximation and Projection)

- Preserves **local + global** structure.
- Scalable and fast.
- Good for semantic visualization of embeddings.

Key params: n_neighbors, min_dist, metric



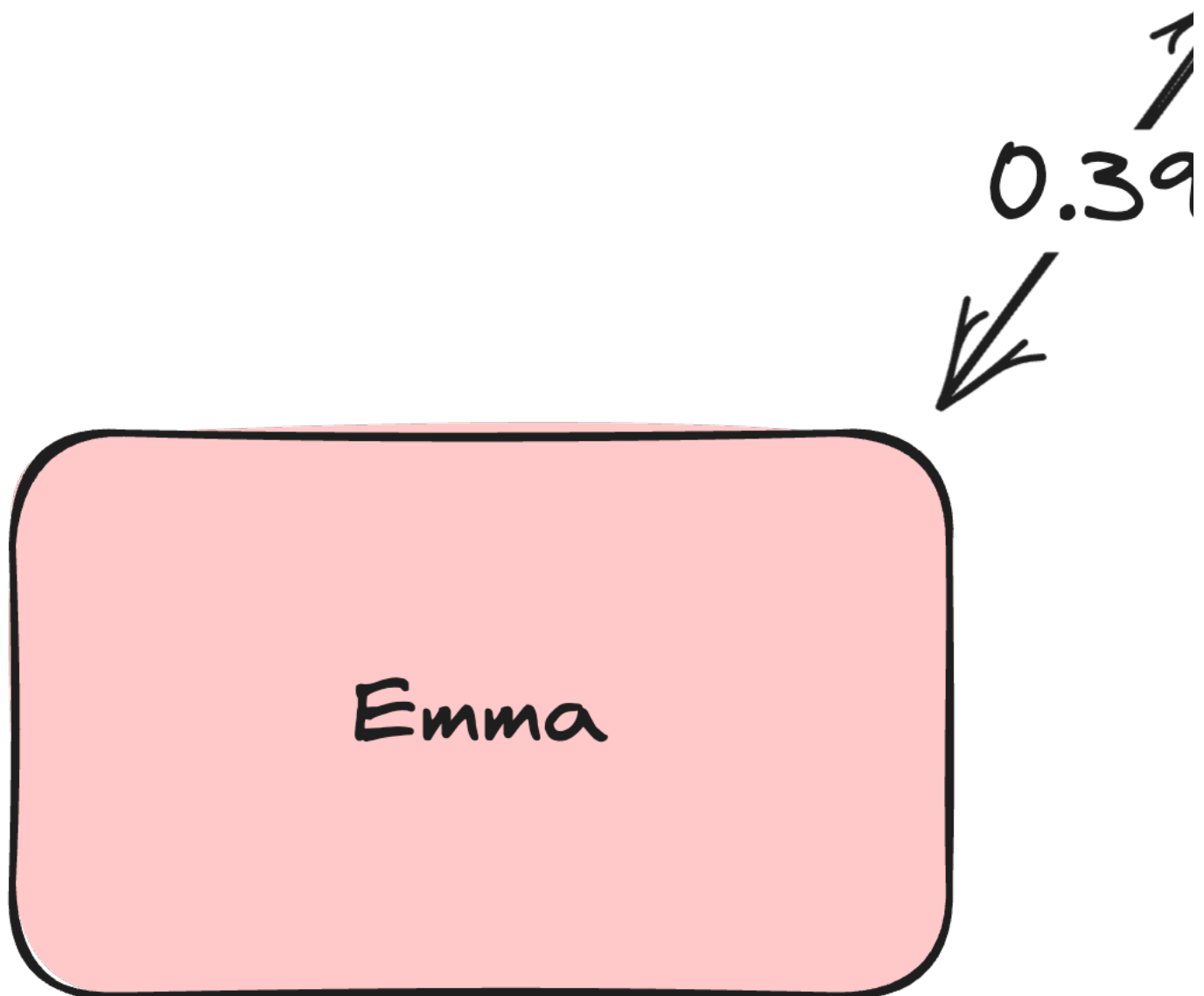
 **Experiment:** Illustrations based on ~2K excerpts of 20Newsgroups, covering 4 topics. Dimensionality reduction methods applied to 384-dimensional embeddings obtained with `all-MiniLM-L6-v2` SentenceTransformer, using cosine distance for MDS and default arguments from `sklearn` and `umap` libraries for PCA, t-SNE, and UMAP.

 **Tip:** Use *PCA* for preprocessing and transformation interpretability, *t-SNE* or *UMAP* for visualization, *MDS* for preserving metric geometry.

For more information about UMAP, see [this blogpost](#) and the mammoth example ([Coenen and Pearce 2019](#)).

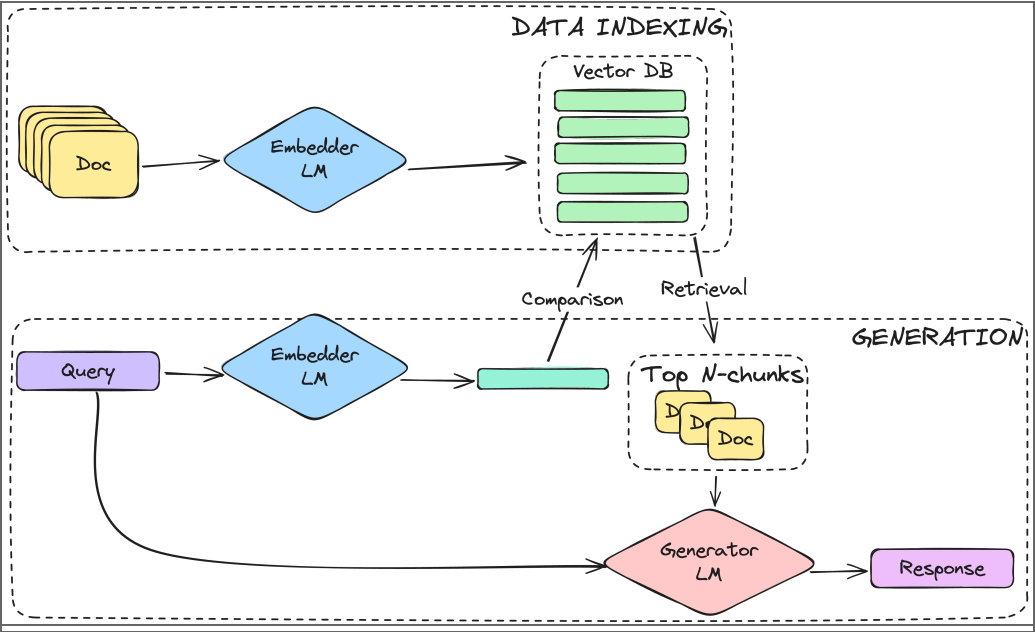
3.3 What can we do now that we have defined distances?

3.3.1 Compare Documents!



*Quick experiment with random chunks from *Pride and Prejudice*, and *Emma* by J. Austen, and *1984* by G. Orwell. Cosine distance computed on embeddings obtained with [all-MiniLM-L6-v2](#) SentenceTransformer.*

3.3.2 Retrieve Documents



Naive Retrieval Architecture Illustration (→ Applied to RAG.)

3.3.3 Group Documents

Overview of common clustering algorithms and their main features.

K-means

- Partitions data into **K clusters** minimizing within-cluster variance.
- Assumes spherical, equally sized clusters.
- Fast, scalable, but sensitive to outliers and initialization.

Key params: `n_clusters`, `init`, `n_init`, `max_iter`

DBSCAN

- Density-based: forms clusters of **closely packed points**.
- Identifies outliers as noise; no need to predefine K.
- Can struggle with varying densities or high-dimensional data.

Key params: `eps`, `min_samples`, `metric`

HDBSCAN

- Hierarchical extension of DBSCAN with **adaptive density**.
- Automatically finds stable clusters and handles noise robustly.
- Excellent for high-dimensional embeddings (e.g., SBERT, UMAP).

Key params: `min_cluster_size`, `min_samples`, `cluster_selection_epsilon`

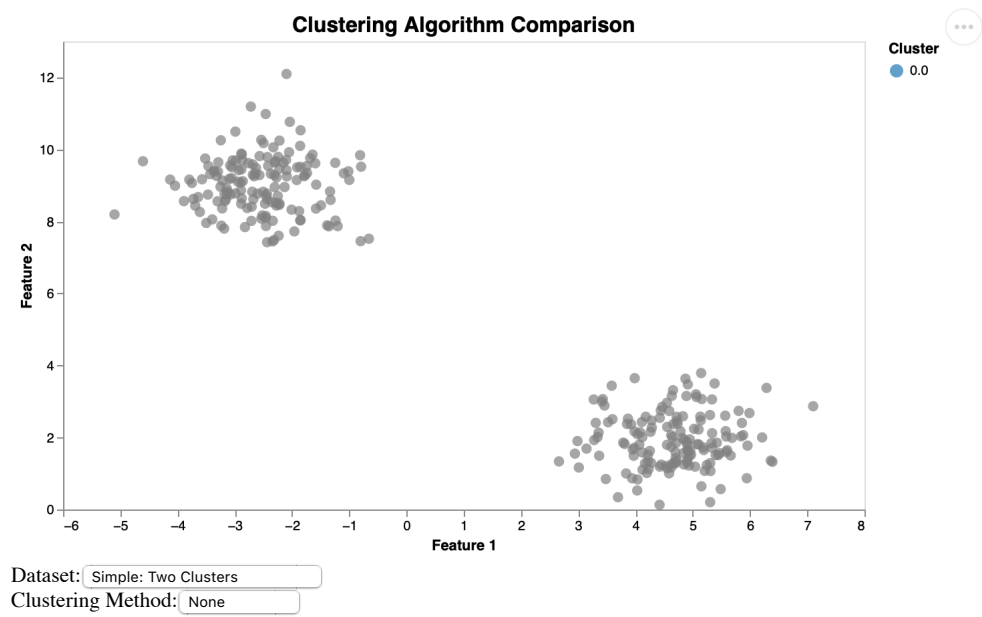
BIRCH

- Incremental clustering using a feature **tree structure**.
- Handles large or streaming datasets efficiently.
- Works best for roughly spherical clusters.

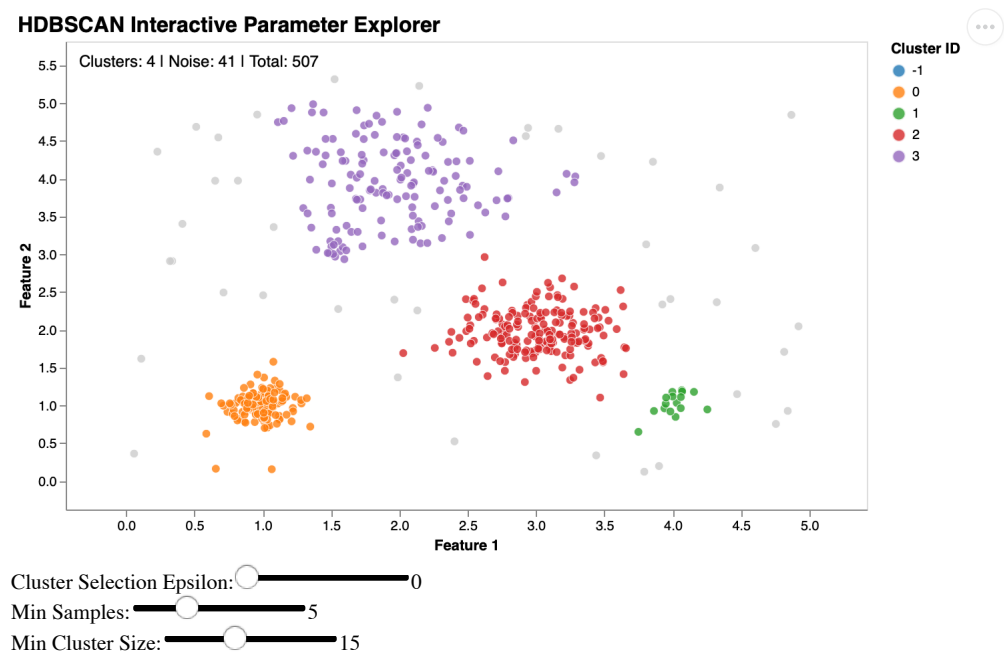
Key params: `threshold`, `branching_factor`, `n_clusters`

💡 **Tip:** Use *K-means* for simple clusters, *DBSCAN/HDBSCAN* for arbitrary shapes and noise, *BIRCH* for large-scale or online data.

3.3.3.1 EXAMPLES



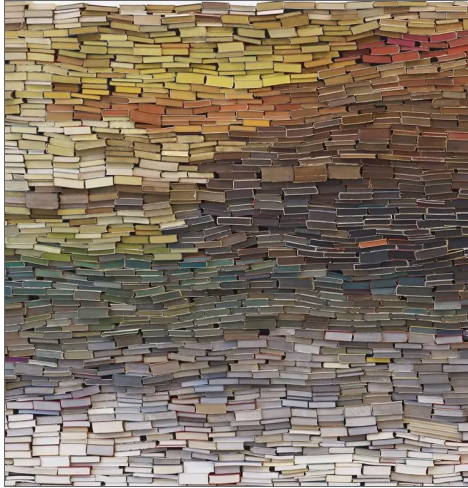
3.3.3.2 HDBSCAN Focus



4 Topic Modeling

4.0.1 Introduction

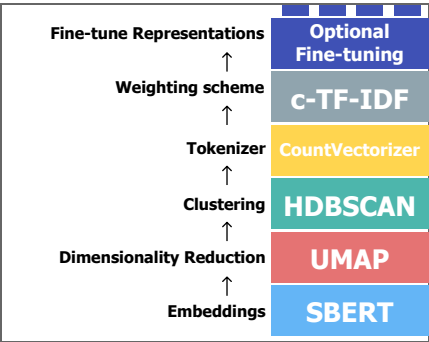
- Topic Model: *unsupervised* algorithm
 - Discover main ‘themes’ in an unstructured corpus
- Multitude of methods
 - “proper” modeling
 - adapted tasks



Enclosed Content Chatting Away In The Colour Invisibilty, 2009, Anouk Kruithof.
(cropped).

4.1 BERTopic

4.1.1 BERTopic Overview



From (Grootendorst 2022)

4.1.2 Documents Clustering









4.1.4 BERTopic Autopsy





4.1.5 BERTopic In Practice

```
1 from bertopic import BERTopic
2 texts = [] # your texts here
3 topic_model = BERTopic()
4
5 topics, probs = topic_model.fit_transform(documents)
```

4.2 Latent Dirichlet Allocation (LDA)

4.2.1 LDA Overview

Latent Dirichlet Allocation (Blei, Ng, and Jordan 2003)

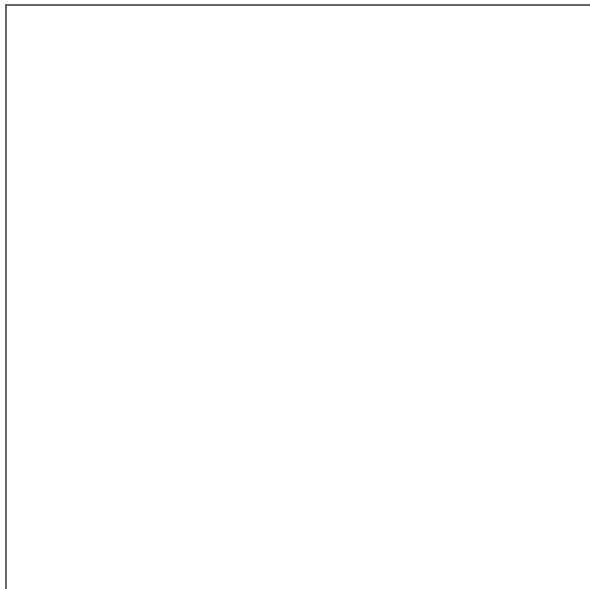
= *proper* TM



LDA Intuition

- **Topic** = Probability **distribution** over **words** from fixed vocabulary.
- **Document** = **Mixture** of corpus-wide **topics**.

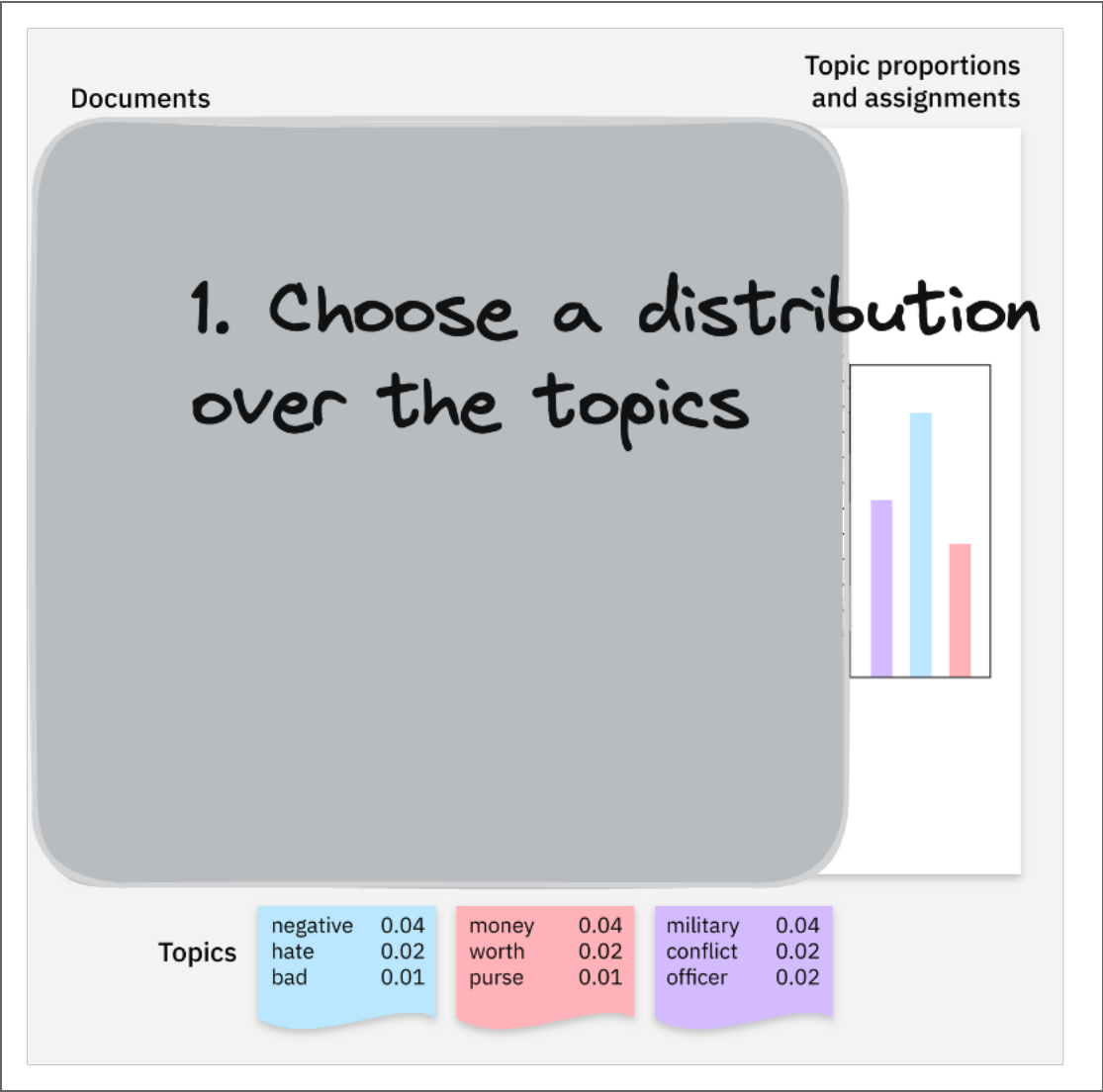
\\(\\to\\) **Generative process**



The intuitions behind latent Dirichlet allocation.

Extracted from (Murel and Kavlakoglu 2025), inspired by (Blei 2011).

4.2.2 LDA: Generative model

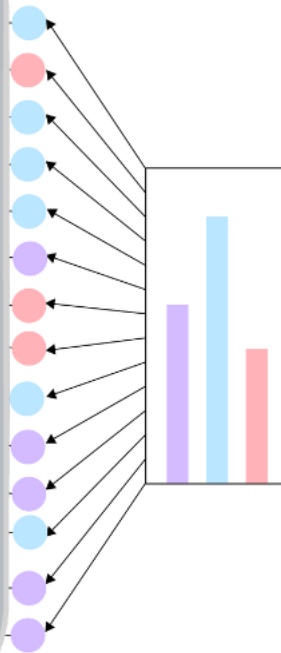


Define doc

Documents

Topic proportions
and assignments

2. For each word,
sample a topic
assignment



Topics

negative	0.04
hate	0.02
bad	0.01

money	0.04
worth	0.02
purse	0.01

military	0.04
conflict	0.02
officer	0.02

> Sample words' topics

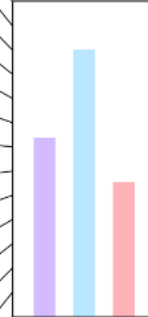
Documents

Shakespeare's *Othello*

RODERIGO: Tush, never tell me, I take it much unkindly

Topic proportions and assignments

3. Sample the word from corresponding topic



Topics

negative	0.04
hate	0.02
bad	0.01

money	0.04
worth	0.02
purse	0.01

military	0.04
conflict	0.02
officer	0.02

> Sample words from topics

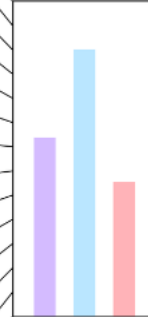
Documents

Topic proportions and assignments

Shakespeare's *Othello*

4. Iterate N times

RODERIGO: Tush, never tell me, I take it much unkindly
 That thou, Iago, who hast had my purse—
 As if the strings were thine, should know of this.
 IAGO: 'Sblood, but you'll not hear me. If ever I did dream
 Of such a matter, abhor me.
 RODERIGO: Thou told'st me
 Thou didst hold him in thy hate.
 IAGO: Despise me
 If I do not. Three great ones of the city,
 In personal suit to make me his lieutenant,
 Off-capped to him; and, by the faith of man,
 I know my price, I am worth no worse a place.
 But he, as loving his own pride and purposes,
 Evades them with a bombast circumstance,
 Horribly stuffed with epithets of war,
 And in conclusion,
 Nonsuits my mediators. For "Certes," says he
 "I have already chose my officer."
 And what was he?
 Forsooth, a great arithmetician,
 One Michael Cassio, a Florentine,
 A fellow almost damnd in a fair wife,
 That never set a squadron on the field,
 Nor the division of a battle knows



Topics

negative 0.04
 hate 0.02
 bad 0.01

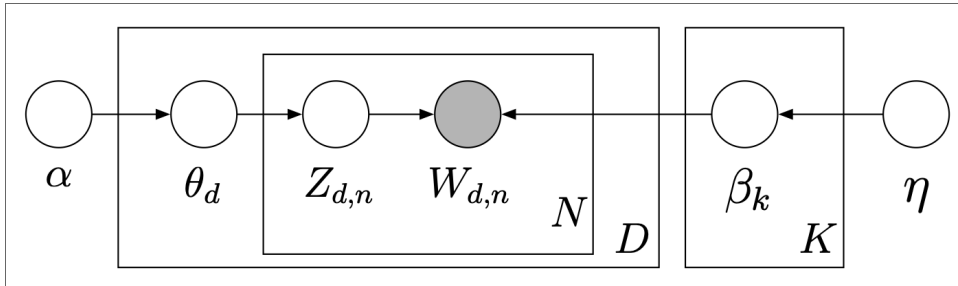
money 0.04
 worth 0.02
 purse 0.01

military 0.04
 conflict 0.02
 officer 0.02

> Iterate

Figure 1: LDA Generative Steps

4.2.3 LDA Graphical Model



LDA Graphical Model. Extracted from (Blei 2011).

Shakespeare's Othello

RODERIGO: Tush, never tell me, I take it much unkindly.
That thou, Iago, who hast had my ~~glove~~,
As if the strings were thine, should know of this.
IAGO: 'Sblood, but you'll not hear me. If ever I did dream
Of such a matter, ~~alight~~ ~~me~~.
RODERIGO: Thou told'st me
Thou didst hold him in thy hate.
IAGO: Despise me:
If I do not, Three great ones of the city,
In personal suit to make me his ~~lieutenant~~,
Off-capped to him; and, by the faith of man,
I know my ~~price~~, ~~I am worth~~ ~~go~~ worse a place.
But he, as loving his own pride and purposes,
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And in conclusion,
Nonsuits my mediators. For "Certes," says he,
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And what was he?
Forsooth, a great arithmetician,
One Michael Cassio, a Florentine,
A fellow almost damnd for a fair wife,
That never set a ~~squadron~~ ~~the~~ field,
Nor the division of a ~~battle~~ knows

Observed State: raw document.

$$\begin{aligned} & \left[p(\beta_{1:K}, \theta_{1:D}, z_{1:D}, w_{1:D}) = \right. \\ & \left. \prod_{i=1}^K p(\beta_i) \prod_{d=1}^D p(\theta_d) \left(\prod_{n=1}^N p(z_{d,n} | \theta_d) p(w_{d,n} | z_{d,n}, \beta_k) \right) \right] \end{aligned}$$

4.2.4 LDA In Practice

```
1 from gensim.utils import simple_preprocess
2 from gensim import corpora
3 from gensim.models import LdaModel
4
5 texts = [] # your texts here (might be a good idea to do a bit of c
6 dictionary = corpora.Dictionary(texts) # extracts vocabulary
7 corpus = [dictionary.doc2bow(text) for text in texts] # documents Bo
8
9 lda_model = LdaModel(
10     corpus=corpus,
11     id2word=dictionary,
12     num_topics=92, # YOUR NUMBER OF TOPICS HERE!
13     random_state=3,
14 )
```

4.3 Summary

4.3.1 TM Comparison

	LDA	BERTopic
Approach	Count-based	Neural
# topics	Pre-determined	Automatically determined
Document	Mixture of topics	One topic
Topic	Distribution over words	Extracted from (disjoint) groups of documents

& other approaches (LSA, variants of LDA, NMF, Fastopic, ...).

For instance see ([Egger and Yu 2022](#)): offers a comparison of diverse TMs applied to Twitter data.

4.4 Evaluation

4.4.1 A hard task!

Reminder: topic models are *unsupervised* (= no ground-truth)

What do we want from a topic model?

Desiderata: *A good TM should...*

1. ... **represent** the **data** well (*statistical fit*)
2. ... provide **coherent topics** (*semantic fit*)

\\to\\) *How to measure that?*

4.4.1.1 STATISTICAL FIT

- Perplexity
 - Assess the ability of a generative process to model observed documents
 - ⚠ Not applicable to BERTopic!
 - (we'll come back to the notion of *perplexity* in a later class)
- Document-topic similarity
 - Compute average similarity between documents and assigned topic embeddings
 - “Native” for BERTopic, not for LDA (could still be applied)
- Topic Coverage / Diversity
 - Proportion of unique words among the top-N words across topics



Caution

For embedding-based models, *statistical fit* is rarely directly evaluated (rather relies on the evaluation of their components).

4.4.1.2 SEMANTIC FIT



Coherence

Topics are good if their top words tend to appear together in similar contexts or documents.

Several operationalization of coherence:

- UMass: $\mathcal{C}_{\mathrm{UMass}} = \log \frac{D(w_i, w_j)}{D(w_i)}$
- UCI: $\mathcal{C}_{\mathrm{UCI}} = \frac{P(w_i, w_j) + 1}{P(w_i)P(w_j)}$
- ...

4.4.2 Importance of human judgement



Automatic evaluation $\neq$ human understanding (of topics)

- Corpus- and parameter-dependent $\rightarrow$ inconsistent comparisons (Newman et al. 2010; Röder, Both, and Hinneburg 2015)
- Misalignment with human judgement (Chang et al. 2009; Hoyle et al. 2021)
- Optimising metrics can harm interpretability (Mimno et al. 2011)

👁️ Human judgement remains essential!

4.5 Example:

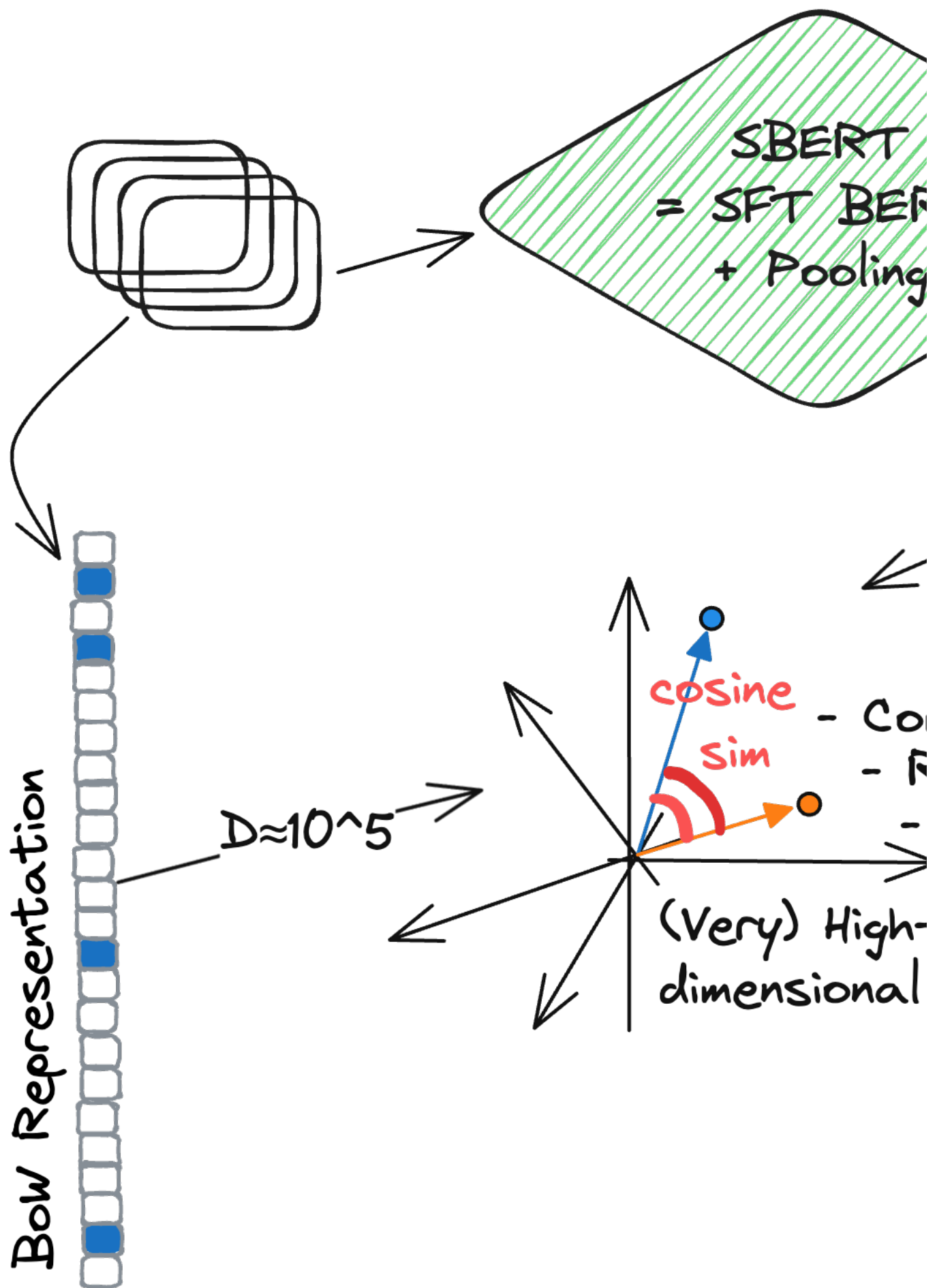
Demographic Representation and Collective Storytelling in the Me Too Twitter Hashtag Activism Movement ([Mueller et al. 2021](#)).

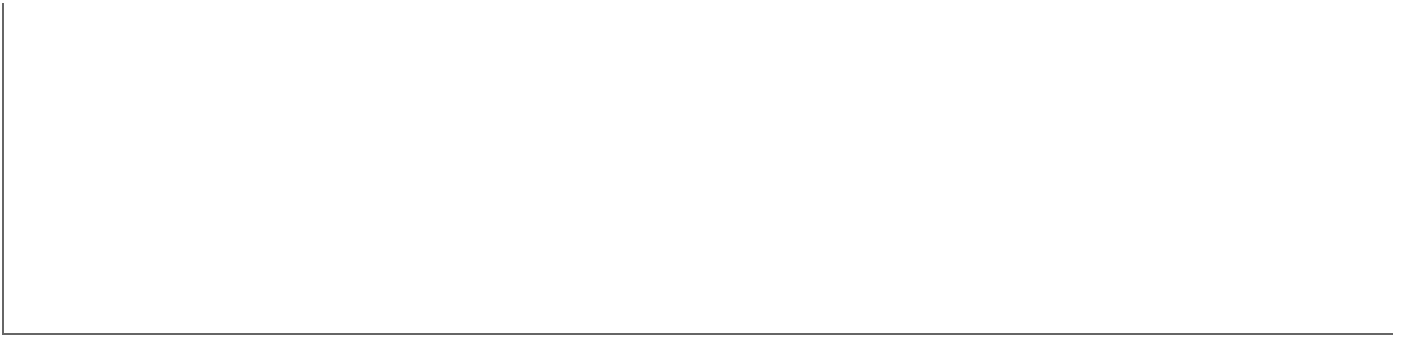
Demographic Representation and Collective Storytelling in the Me Too Twitter Hashtag Activism Movement

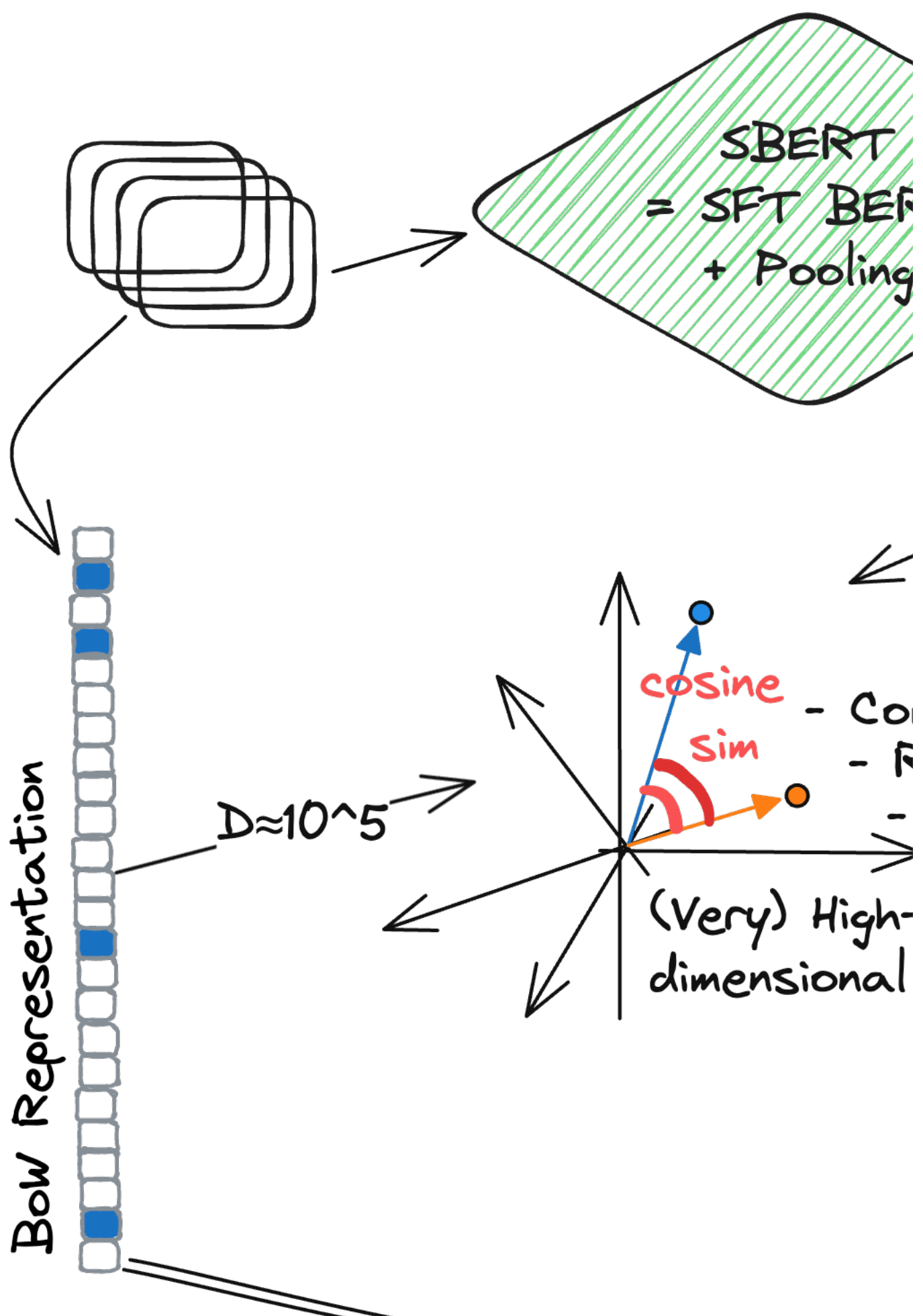
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The #MeToo movement on Twitter has drawn attention to the pervasive nature of sexual harassment and violence. While #MeToo has been praised for providing support for self-disclosures of harassment or violence and shifting societal response, it has also been criticized for exemplifying how women of color have been discounted for their historical contributions to and excluded from feminist movements. Through an analysis of over 600,000 tweets from over 256,000 unique users, we examine online #MeToo conversations across gender and racial/ethnic identities and the topics that each demographic emphasized. We found that tweets authored by white women were overrepresented in the movement compared to other demographics, aligning with criticism of unequal representation. We found that intersected identities contributed differing narratives to frame the movement, co-opted the movement to raise visibility in parallel ongoing movements, employed the same hashtags both critically and supportively, and revived and created new hashtags in response to pivotal moments. Notably, tweets authored by black women often expressed emotional support and were critical about differential treatment in the justice system and by police. In comparison, tweets authored by white women and men often highlighted sexual harassment and violence by public figures and weaved in more general political discussions. We discuss the implications of work for digital activism research and

5 Takeaways









6 Hands-on

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[fbclid=IwAR0ribQrR3TwrKb0rOWW6gsvE1CuoxIfcPizvw_ReENZoQMJ229o5Ar2u](https://dl.acm.org/doi/abs/10.1145/2347736.2347755?fbclid=IwAR0ribQrR3TwrKb0rOWW6gsvE1CuoxIfcPizvw_ReENZoQMJ229o5Ar2u)
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